

Haseeb Asif

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Summary

Research oriented Computer Science undergraduate with experience in deep learning, generative AI, large language models, and agentic systems. Skilled in implementing and reproducing machine learning methods, conducting experimental evaluation, and developing AI applications. Currently exploring prompt injection resilience and AI safety for large language models, with research interests in post training, reinforcement learning, alignment, and trustworthy AI systems.

Education

FAST National University, B.S in Computer Science | Karachi, Pakistan | CGPA: 3.51/4

Aug 2023–Present

Achievements & Awards: 4x Deans List | ACM Automation (Best Team Award)

Relevant Coursework: Linear Algebra, Calculus, Multivariable Calculus, Probability & Statistics, Discrete Structures, Data Structures, Recommender Systems, Information Retrieval, Artificial Intelligence

Experience

Algoma University, Mitacs Globalink Research Intern | Sault Ste. Marie, Canada

July 2026–Current

- Conducting research on visual similarity based product recommendation systems, benchmarking embedding based architectures across CNN feature extractors (ResNet, VBPR) and vision language models (CLIP, FashionCLIP) for cross modal retrieval.
- Architecting a scalable experimentation pipeline on the Amazon Review Data 2018 (Clothing, Shoes & Jewelry) corpus to evaluate visual embedding techniques against retrieval accuracy and recommendation relevance metrics.
- Synthesizing a comparative literature review spanning CNN and VLM based recommendation architectures to surface state of the art techniques and identify gaps driving the next phase of experimentation.

Stevens Institute of Technology, Research Assistant | Remote (United States)

Apr 2025–May 2025

- Developed a custom convolutional neural network in PyTorch with residual connections and dropout, achieving 95.2% classification accuracy on 5,000+ anonymized brain MRI scans for neurological disorder detection.
- Built a reproducible preprocessing pipeline in Python (NumPy, SciPy, Pillow) with intensity normalization, skull stripping, and affine transformations to standardize heterogeneous MRI datasets.
- Ran large-scale hyperparameter optimization with Ray Tune (distributed grid, random, and Bayesian search) and early stopping, reducing training time by 30%.
- Applied SHAP and LIME to produce voxel-level saliency maps for clinical validation and peer-reviewed research.

PROCOT, Deputy Head Automation | Remote (Karachi, Pakistan)

Jan 2025–Feb 2025

- Designed and implemented an end-to-end certificate automation pipeline using FastAPI, MongoDB, and Email MIME, dynamically generating and tailoring certificates with unique UUID URLs for verification.
- Developed secure and scalable dynamic endpoints using FastAPI and MongoDB for certificate validation.
- Deployed the pipeline on AWS Cloud, optimizing scalability and performance.
- Tested and validated endpoints with Postman Agent.

Skills

Programming Languages: Python, C, C++, JavaScript | ML Frameworks: Scikit-learn, TensorFlow, PyTorch, PyTorch Lightning, JAX, ONNX, MLflow, Ray, Apache Airflow, Weights & Biases, TorchServe | Computer Vision: OpenCV, CARLA, Matplotlib, Plotly | Geospatial & Remote Sensing: Google Earth Engine, Sentinel-2 MSI, NDVI/NDBI/LST Analysis, Mapbox GL, GeoJSON | Data Processing: Pandas, NumPy | LLM & RAG: LangChain, Pinecone, FAISS, ChromaDB, CrewAI, Model Context Protocol (MCP) | Web & Automation: FastAPI, Streamlit, Selenium, Postman, Discord.py | Frontend Development: Next.js, HTML, CSS | DevOps: Git, GitHub Actions, Docker | Tools: VS Code, CursorAI, Google Colab, Hugging Face Spaces | Databases: MongoDB | Soft Skills: Research-Oriented Thinking, Problem Solving, Scientific Communication, Team Collaboration, Self Directed Learning

Projects

EarthLink AI — EnviroCast Global Engineering Outlook Hackathon | 1st Place Overall

Feb 2026

- Architected an AI powered geospatial intelligence platform transforming natural language queries into actionable environmental insights using Sentinel-2 satellite imagery, agentic LLM workflows, and dynamic tool orchestration via FastAPI and Google Gemini.
- Engineered a multi-step environmental analysis pipeline computing NDVI, land surface temperature (LST), built up intensity (NDBI), elevation, and vegetation metrics from multispectral satellite data for climate resilience and urban sustainability analysis.
- Developed an interactive full-stack geospatial visualization interface using Next.js, React, and Mapbox GL, enabling real time rendering of environmental hotspots, urban heat islands, and satellite-driven regional comparisons.
- Built scalable architecture designed for live Google Earth Engine integration, supporting large-scale geospatial workflows for environmental equity analysis and climate intelligence accessible through conversational prompts.

Image2Depth

July 2025

- Implemented a Pix2Pix based image-to-image translation system in PyTorch to estimate monocular depth maps from single-view dashboard RGB images, leveraging a U-Net generator with skip connections and a PatchGAN discriminator for enforcing local realism.
- Optimized training with a composite objective of adversarial loss and L1 pixel-wise reconstruction loss, achieving 23.8% reduction in MAE and significantly sharper depth boundaries compared to baseline CNN regressors.
- Engineered a robust preprocessing pipeline to parse and align RGB-depth training pairs while preserving spatial consistency and semantic structure across domains.
- Automated hyperparameter tuning with Ray Tune (ASHA scheduler) and integrated Weights & Biases with MLflow for experiment tracking and reproducible benchmarking, achieving 31% faster convergence across training runs.

DeepFaceEdit

June 2025 – July 2025

- Architected and engineered a full featured desktop application for real time human face synthesis and latent space exploration, leveraging StyleGAN2-ADA with a PyTorch backend.
- Implemented a latent space manipulation engine enabling fine grained semantic editing of facial attributes through learned direction vectors in the high dimensional W^+ latent space.
- Engineered an advanced image projection module to map real photographs into StyleGAN's latent space by optimizing latent vectors, achieving 18% lower L2 reconstruction error compared to baseline pipelines.
- Integrated dlib based facial landmark detection for precise face alignment prior to projection, improving latent correspondence and perceptual fidelity of reconstructed identities.

Advance Lane Detection (Classical CV)

Apr 2025 – May 2025

- Developed a complete lane detection pipeline from scratch using OpenCV and NumPy, tailored for ADAS style real time video feeds and static road imagery.
- Implemented camera calibration, distortion correction, and perspective transform (bird's eye view) to reduce geometric distortions and simplify lane geometry.
- Designed a binary thresholding and sliding window search strategy to segment lane pixels and fit second order polynomials, enabling robust detection on straight and curved roads.
- Calculated lane curvature and vehicle offset in real world units by mapping image space to meter scale, validating the system with 94.2% pixel-level accuracy under challenging driving conditions.

Hackathon Projects

EarthLink AI — EnviroCast Global Engineering Outlook: Furthering the Future | 1st Place

Feb 2026

Overall

- Built an agentic geospatial AI platform converting natural language prompts into real time satellite driven environmental insights using Sentinel-2 imagery, NDVI, LST, and NDBI metrics.
- Won 1st Place Overall among all submissions for innovation in AI powered environmental intelligence and climate data accessibility.

EarthLink AI — ORIGIN Hackathon | Best Social or Environmental Contribution

Feb 2026

- Recognized for outstanding social and environmental impact through EarthLink AI's satellite driven geospatial intelligence platform, which democratizes access to climate and environmental data through natural language interaction.

Additional Projects & Paper Reproductions

- Built a Load Forecasting system using LSTM for electricity demand prediction.
- Built NeRF (Minimal PyTorch), reimplementing Neural Radiance Fields for 3D view synthesis.
- Created XRayGAN for synthetic chest X-ray generation, achieving 89.3% ResNet classifiability.
- Trained a Pix2Pix GAN model for Satellite-Imagery to Map Translation, achieving SSIM 0.82.
- Built Face Reconstruction using Scikit-learn, PCA, and tree based regressors to predict missing facial structure.
- Reproduced Knowledge Distillation in TensorFlow, achieving compact student models with minimal performance drop.
- Developed NeuroInk, a neural style transfer tool using VGG19.
- Automated certificate distribution for Coder's Cup 24, generating and emailing 1000+ personalized certificates.

Certificates

ML Summer School 2025 — Cohere

Participated in a curated program focused on foundational and advanced machine learning concepts.

2025

Generative Adversarial Networks (GANs) Specialization — DeepLearning.AI

Gained hands-on experience building and training GAN models for image generation tasks.

2025

Ray Fundamentals Accreditation — AnyScale

Learned fundamentals of distributed computing and ML training using Ray.

2025

Deep Learning with PyTorch: Siamese Network — Coursera

Completed a course on Siamese Networks using PyTorch for similarity learning tasks.

2025